

The physical internet

CMS.100

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Assignment 3

In this paper, **you will choose a media topic to systematically gather data about, code (categorize) that data in some way, visualize the data, and analyze your findings.** Instead of a platform, you can also analyze some media phenomenon.

You can gather data about this phenomenon through any of the qualitative (interviews, observations), quantitative, or experimental methods that you've learned about this semester.

A3.1 project pitches

Wed Oct 7	Fri Oct 9
Alicia Ait-Azzouzene	Trish Zheng
Tan M. Le	Olumoroti Anibaba
Jing Lin	Joseph Castro
Tri H. Nguyen	Amy W. Chen
Axel F. Nunez	Sean K. Harrison
Megna Pillarisetti	Rudra Pethani
Mingjia Zhang	Ryan C. Ruiz
Eric Zheng	

From Wednesday

“Fiber-optic cables for internet transmissions follow many of the same routes as the original telegraph lines.”

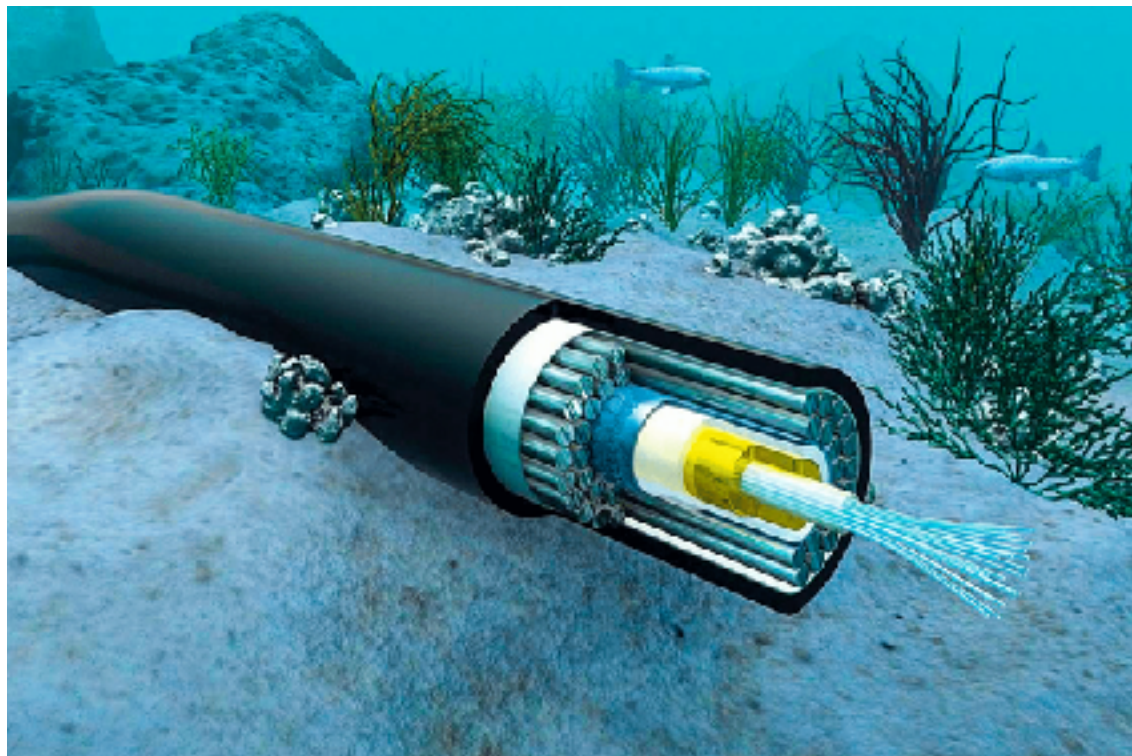
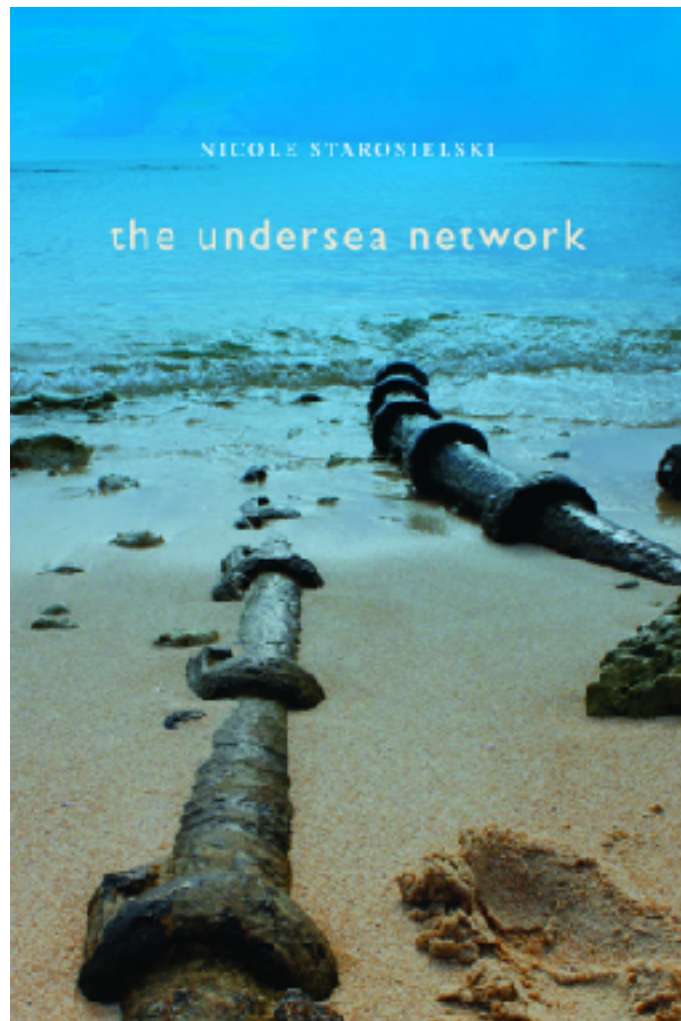
Gitelman and Mullaney (2021), p. 178

Why? Today's reading explains.

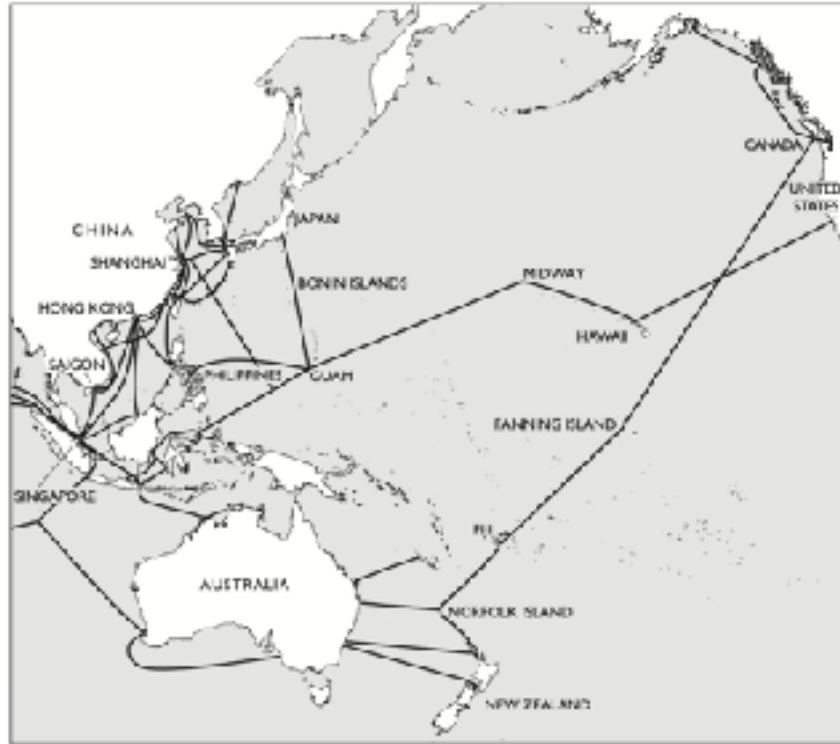


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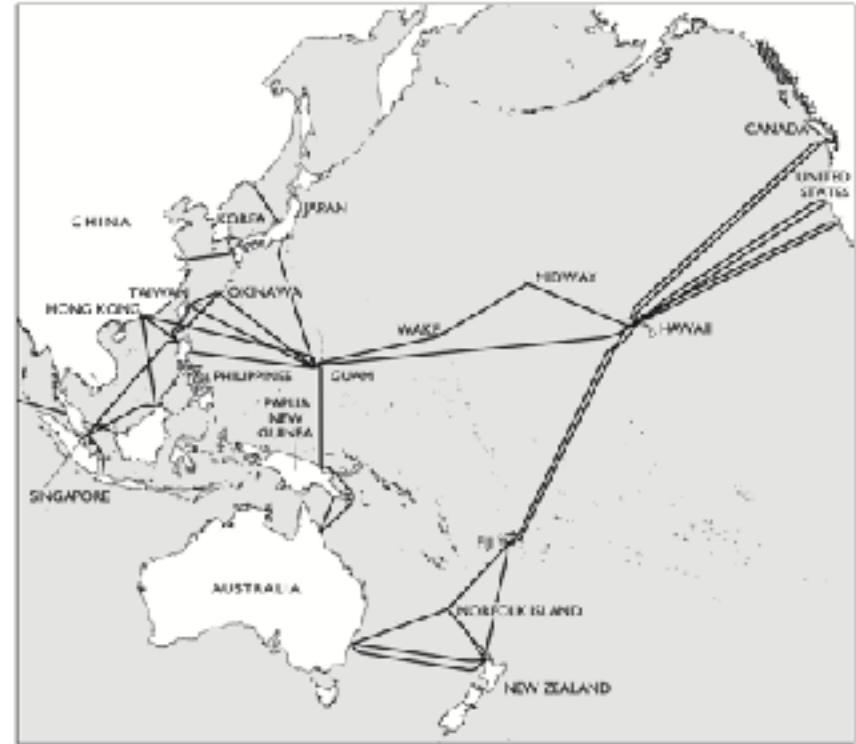
the undersea network



How does the internet work? (a prehistory)



MAP 1.1. Transpacific cable routes, 1922.

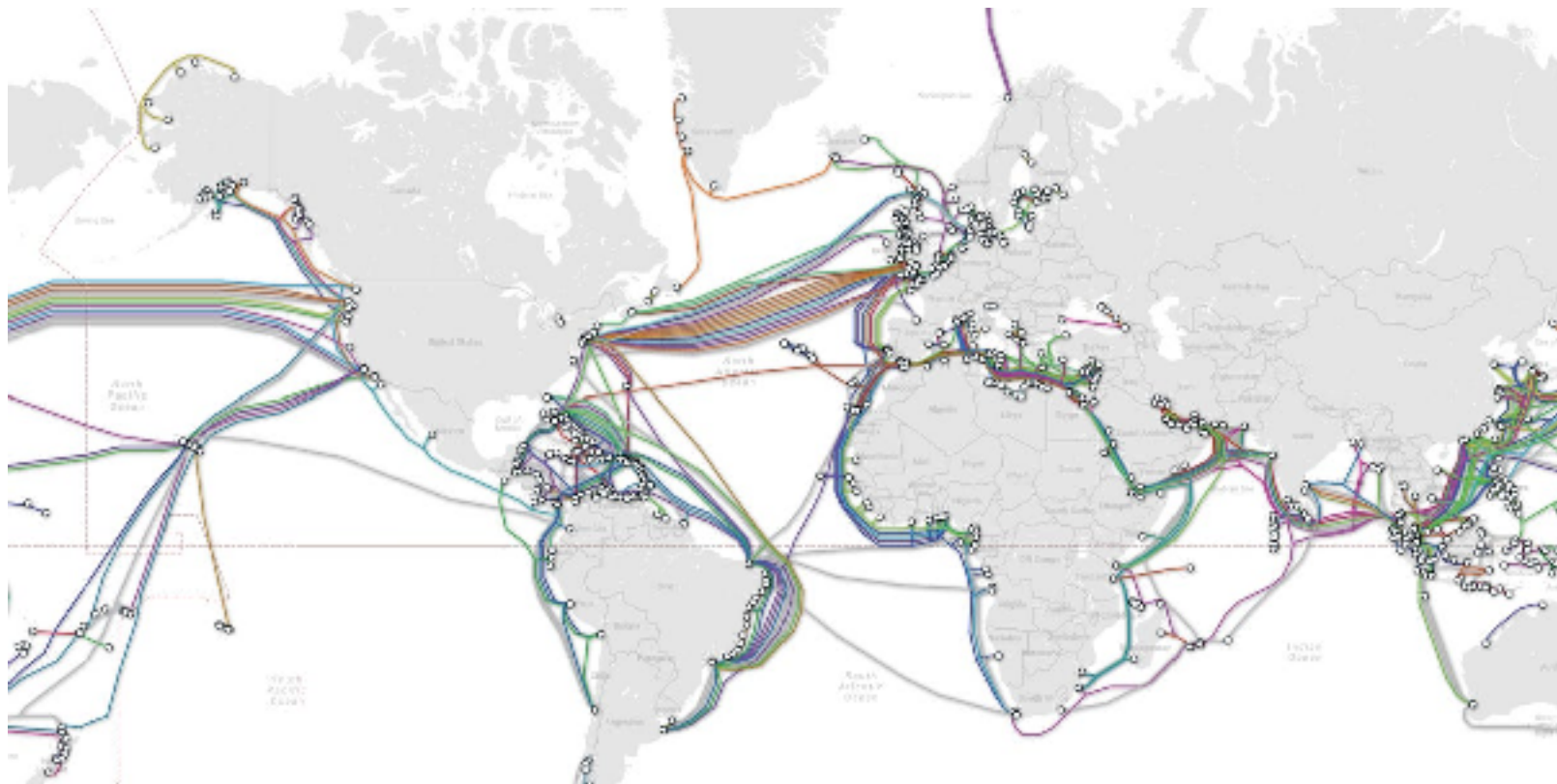


MAP 1.2. Transpacific cable routes, 1982.

EASTERN TELEGRAPH C^d SYSTEM AND ITS GENERAL CONNECTIONS.



Undersea network cables





③
INTERNET BACKBONE

②
INTERNE

The network is...

wired

rather than wireless

semicentralized

rather than distributed

territorially entrenched

rather than deterritorialized

precarious

rather than resilient

rural and aquatic

rather than urban

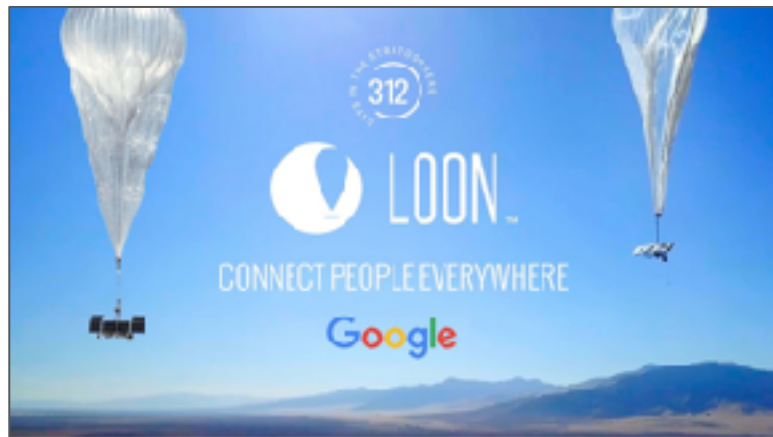
Project Loon (Google)

Digital Silk Road (China)

IDEAS

The real reason China is pushing “digital sovereignty” in Africa

As the Chinese “tech stack” leads from undersea cables to smartphones and fintech apps, concerns grow for the digital future of ordinary Africans.



FEATURES

Sea change

Google and Meta's new subsea cables mark a tectonic shift in how the internet works, and who controls it.

By ANDREW BLUM and CAREY BARAKA

10 MAY 2022

Double-edged sword of “increased internet access”

POLICY / MOBILE / US & WORLD

Facebook's Free Basics service has been banned in India / The country has said no to zero-rating schemes

By [James Vincent](#), a senior reporter who has covered AI, robotics, and more for eight years at The Verge.

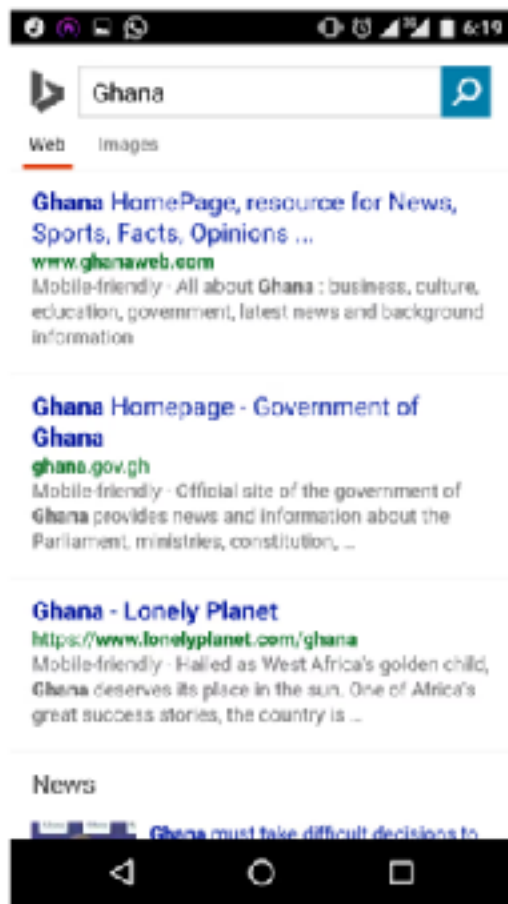
Source [Telecom Regulatory Authority of India](#) | Via [BBC News](#)

Feb 8, 2018, 6:22 AM EST

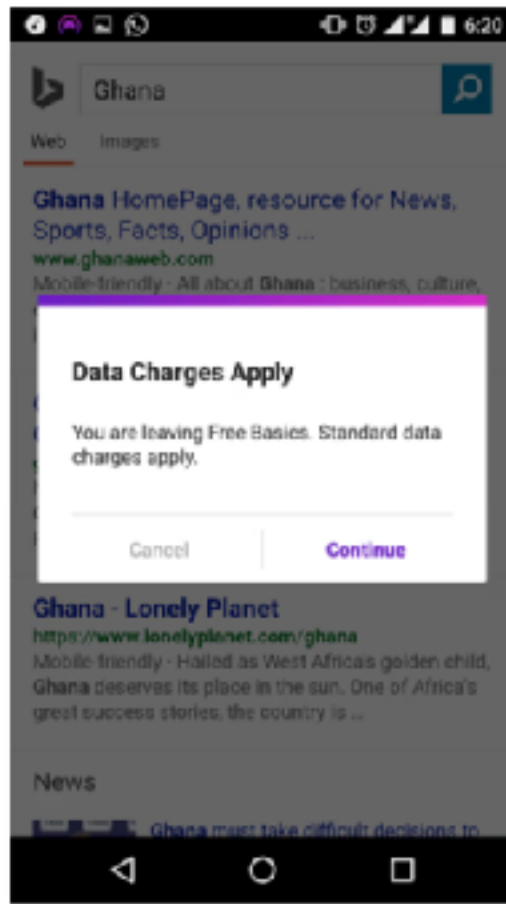
THE WALL STREET JOURNAL

Facebook Promised Poor Countries Free Internet. People Got Charged Anyway.

Its global program, designed to boost growth outside the U.S., leads to unexpected data costs; the company blames software glitches

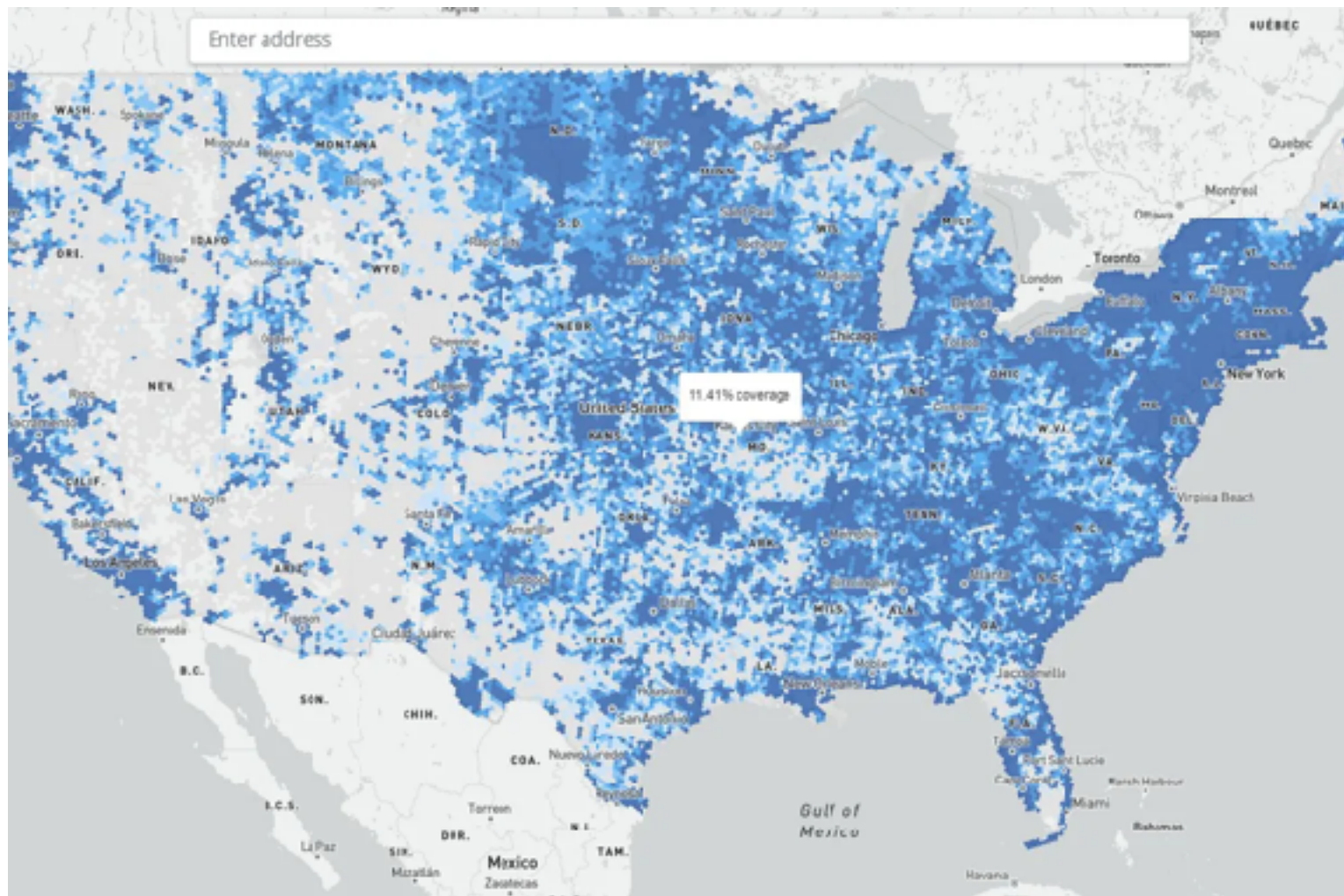


A Bing search in Ghana



After selecting a link, the user is told that standard data charges will apply

Enter address



HANK VS. HANK

NET NEUTRALITY

EXPLAINED



Other environmental impacts...

- Electromagnetic fields (EMF) generated by submarine power cables may affect marine life navigational abilities (area of future study)
 - Creates new marine ecosystems (opportunistic species) but these form “climax communities”
 - See Ed Yong’s *A Vast World* for more!
- Huge impacts during cable burial, risk of entanglement, introduction of artificial substrates (long-term damage)
 - Cables could however minimize fossil fuel consumption to connect onshore data centers to renewable grids

Part 1: Two maps of the same network

submarinecablemap.com

How many cables land in your place?
Who owns the newest one?

surfacing.in

Find one landing site. What does it show
that the cable map does not?

Then: which of her five claims does your place support or complicate?

Before you go

Hand in your sheet (annotations + A1 sentence)

A3.1 pitches: Oct 7 and 9. Add one slide to the shared deck.

A1 due: 9 am, Fri Oct 16